

OSW26BZ06-NV028 - 5G Signature Tracking Mitigation via Cyber Deception

OSD FutureG | SBIR | Release 6 | Opens Sep 23, 2026 (TPOC direct contact ends) | DSIP Q&A closes Oct 7, 12:00 ET | Proposals due Oct 21, 12:00 ET

Recommendation: PURSUE if a cellular-protocol partner is available. Phase I is a concept design, feasibility study and design document; the hard problem (AI personas that defeat ML pattern-of-life tracking) is an ML problem. Phase II needs small decoy devices. ITAR. Confidence MEDIUM.

Phase I award	Up to \$323,090, 6 months (OSW FutureG cost volume rule)
Proposal format	Technical volume 20 pages
Compliance	ITAR/EAR clause. CMMC Level 2 (Self).
Phase I deliverables	Kickoff/TIM slides, monthly status reports, Preliminary Design Document (AI-persona generation and ID-swapping mechanics), Phase II plan, final report
Phase II	Prototype to TRL 7: mission-management software plus hardened decoy boxes (router-sized team device, puck-sized personal device); field demo on a 5G test network

What they want

[TOPIC TEXT] "an advanced digital signature management and deception system that obfuscates cellular metadata ('digital exhaust') to protect personnel, small units, and critical facilities from adversarial surveillance, tracking, and pattern-of-life analysis on 3rd-party commercial networks."

Suggested key performance metrics: 2 (threshold) to 4+ (objective) AI personas broadcast per decoy device; 12 to 24-36 hours battery; under 2 lb (~1 lb objective); obfuscation realism that defeats basic heuristics (threshold) up to advanced ML-based behavioral tracking (objective). Proposers must define their own quantifiable KPMs.

Phase I is six months of concept, algorithms and a software-architecture design document, plus a PDR.

How we would build it

- **Persona generator:** a generative sequence model of mobility and device-usage patterns (location trace, call/data cadence, app-traffic mix, dwell times) conditioned on a mission plan, producing several concurrent believable personas.
- **Adversary model:** our own pattern-of-life tracker (the kind an adversary would run on CDR/location metadata) used as the discriminator - realism is measured as the tracker's failure to separate real from decoy or to find the unit.
- **Mission planner:** AI-enabled planning of when/where personas appear and how identities rotate so deviations that signal 'VIP visit' or 'mission prep' are masked.
- **Device architecture (design only in Phase I):** COTS cellular modems / SDR modules on an SBC with SIM/eSIM identity management; ID-swapping mechanics documented with the legal/operational envelope.
- **Evaluation harness:** synthetic and public mobility datasets; metrics per KPM.

Resources we would use

- Public mobility/trajectory datasets (GeoLife, taxi/GPS traces, open CDR-style research datasets) to train and test persona realism.
- Open-source cellular stacks (srsRAN, OpenAirInterface) and a lab 5G core for Phase II test-network work.
- DoDI 8520.02 (cited) for identity/credential handling constraints.

- Cellular-protocol partner or consultant for identity, SIM and network-attach behavior.
- COTS modems, eSIM management, small SBCs - benchtop-sourceable.

Bottlenecks and risks

- **Protocol and legal depth:** generating decoy identities on third-party networks touches SIM provisioning, IMSI/IMEI behavior and carrier terms; the design document must be credible on this without in-house telecom experience.
- **Realism bar is high:** defeating ML-based behavioral tracking is an adversarial-ML arms race; the honest framing is measurable degradation of tracker performance, not invisibility.
- **Phase II hardware:** hardened 1-lb devices with 24-36 h battery is a product-engineering job; plan a hardware sub.
- **ITAR staffing.**
- **Operational validation:** field demo on a 5G test network requires range access the government does not promise.

Decision factors

- Fit to team: the realism/generation/evaluation core is ML and backend - strong; radios are not.
- Dual use: [TOPIC TEXT] protecting corporate executives, personnel and IP from commercial espionage and tracking - a plausible executive-protection / travel-security product.
- Decisive pre-proposal experiment: train a persona generator on a public trajectory dataset and show a standard pattern-of-life classifier cannot separate synthetic from real traces above chance - directly evidences the 'obfuscation realism' KPM.

Before Oct 7 (Q&A): ask whether Phase I may remain software/design-only, whether a test 5G network will be available in Phase II, and how the sponsor expects identity swapping to be lawful on third-party networks.

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